

Kino-VLA: Closed-Loop Embodied Reflection for Semantic Failure Recovery in Quadrupedal Locomotion

Anonymous Author(s)

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Affiliation and contact details omitted for anonymity

Abstract—Learned controllers and rapid motor adaptation let legged robots reject a wide range of disturbances by reacting continuously to proprioceptive history. A distinct class of physical failures, however, cannot be recovered this way: escaping an elastic tether requires backing off rather than applying more torque, and a foot held by an adhesive trap is proprioceptively indistinguishable from one sinking into mud, yet the two demand opposite recoveries. Such failures require discrete, topological, strategy-level decisions that lie outside the hypothesis space of any continuous adaptive controller. We present Kino-VLA, a closed-loop embodied-reflection framework that couples high-frequency dynamics sensing with a vision-language recovery planner for a quadrupedal robot. Recovery is organized into five layers: a 1 kHz monitor-and-reflex loop that keeps the robot upright while reasoning proceeds; a Kino-Tokens extractor that distills 1 kHz proprioception into the language model’s embedding space under privileged-physics supervision; a vision-language planner that performs causal failure attribution, selects from an expanded recovery-primitive library, and maintains a persistent semantic traversability map; and a control-barrier-function safety shield that projects every proposed primitive onto a provably safe set, with a forward-invariance guarantee that, under a reduced-order model, the closed loop never leaves that set regardless of what the language model outputs. We further contribute Kino-Fail, a benchmark of eleven continuously parameterized failure operators with privileged ground truth and constructively matched attribution-ambiguity pairs, and a privileged-grounded hindsight chain-of-thought data pipeline whose truth-consistency filter rejects any annotation contradicted by simulator physics. In high-fidelity simulation on a physically simulated Unitree Go2, the trained planner attains 0.97 attribution accuracy on semantic-intervention failures, against 0.24 for a tuned rule-based state machine, while the safety shield solves in under 0.2 ms.

Index Terms—Legged robots, vision-language-action models, failure recovery, control barrier functions, embodied AI, safe learning, quadrupedal locomotion.

I. INTRODUCTION

Reinforcement learning has made quadrupedal locomotion remarkably robust. Policies trained in simulation with domain randomization, and augmented with online motor adaptation, cross rubble, mud, and ice and reject pushes that would topple a hand-tuned controller [1]–[4]. This robustness rests on continuous reaction: from a window of proprioceptive history the controller implicitly infers the local dynamics (friction, payload, terrain compliance) and adjusts its gait within a continuous parameter space [3]. For a broad family of disturbances this is the correct response, and we claim no improvement on it.

A separate family of failures defeats continuous reaction. A leg caught by an elastic tether is the canonical case: an adaptive controller raises torque to hold its commanded velocity, the tether stores that energy like a drawn bow, and the accumulated load eventually flings the robot off balance, so the safe action is the counter-intuitive one of backing off. Sudden resistance at a foot is a second case: if the cause is an adhesive trap the robot must step back, label the patch untraversable, and route around it, whereas if the cause is mud it must raise its gait and drive through. The recoveries are opposite, yet the two causes can be tuned to produce identical proprioceptive signatures, matched tangential-resistance curves, slip distributions, and torque spectra, so a controller reading proprioception alone cannot, even in principle, tell them apart.

We make this structure explicit as a *recoverability dichotomy*. A-class failures (uniform low friction, a static load offset, a mildly viscous floor) are recoverable by continuous adaptation, and an adaptive controller should handle them as well as any semantic system does. B-class failures require a discrete, topological, or strategy-level decision (back off, declare an entire region impassable, or abandon a torque-saturated task and request help), and an incorrect causal attribution produces the wrong, frequently opposite, recovery. What sets B-class failures apart is that the decision turns on a cause proprioception cannot resolve but vision can, such as the yellow sheen of a glue board against the brown of mud. These decisions lie outside the hypothesis space of any continuous adaptive controller and inside that of a semantic one. The scientific question we pose and answer empirically follows: which physical failures are recoverable by low-level adaptation, and which require semantic intervention?

Existing learning systems leave the B-class gap open. End-to-end vision-language-action models map observations to actions open-loop and at low frequency, without a force-feedback channel, and commit to a planned action through a developing failure [5]–[8]. Visually closed-loop planners react to RGB at camera rate [9], [10] and stay blind to a friction collapse or an actuator saturation until it has produced an overt visual symptom. Legged navigation models built on language backbones treat locomotion as a one-way black-box skill [11]–[13]: the planner issues velocity or waypoint commands but receives no upward report that the body is slipping or pinned. Closest to our aim, failure-reasoning systems such as RE-

FLECT [14] and AHA [15] summarize multimodal experience to explain and correct manipulation failures, but they attribute after the fact from low-frequency vision, address fixed-base manipulators rather than an underactuated floating-base robot, and inject conclusions as discrete text; on a legged platform they additionally lack the millisecond reflex that would keep the robot upright during the roughly one second of language-model inference.

We present Kino-VLA, a closed-loop embodied-reflection framework that supplies the missing upward channel from 1 kHz dynamics to a semantic planner and the missing downward guarantee that the planner cannot destabilize the robot. Kino-VLA organizes recovery into five layers. A 1 kHz Kino-Monitor computes slip and tracking-error statistics and triggers a reflex loop that stabilizes the body (brake, stiffen, lower the center of mass) within milliseconds, bypassing the language model. A Kino-Tokens extractor compresses a 500 ms proprioceptive window with a 1D convolutional network and a Perceiver Resampler [16], [17] into fixed-length tokens projected into the language model’s embedding space; the latent is supervised by distillation against privileged simulator physics [3], [18], with an auxiliary physics-text contrastive objective [19], so the tokens carry interpretable physical meaning and a calibrated out-of-distribution score. A roughly 1 Hz planner, a LoRA-adapted [20] vision-language model [21], [22], fuses vision, text, and Kino-Tokens to attribute the failure causally, selects a single primitive from an expanded recovery library, and maintains a persistent semantic traversability map onto which physical evidence is written. A control-barrier-function (CBF) safety shield and primitive compiler then project every proposed primitive onto a provably safe set before execution. Two ideas carry the design. First, *the language model proposes, the barrier function disposes*: a CBF quadratic program minimally edits each primitive so the capture point cannot leave a contracted support polygon, which yields a forward-invariance guarantee, under a reduced-order model, that the closed loop remains in its safe set whatever the language model outputs [23], [24]. Second, privileged physics grounds both the latent and the training data, replacing an unanchored black box with quantities the simulator can verify.

To study the dichotomy we introduce Kino-Fail, a benchmark of eleven continuously parameterized failure operators across four simulator mechanism axes, each exposing its parameter vector as privileged ground truth. Three operator pairs are constructed so their proprioceptive statistics coincide, making a proprioception-only baseline fail on them by design, and a dedicated suite sweeps an operator’s parameter across the A/B frontier, turning the question of when semantic intervention becomes necessary into a measurable curve. To train the planner we build a privileged-grounded hindsight chain-of-thought [25] pipeline whose central component is an automated truth-consistency filter: an oracle annotation is kept only if its physical attribution matches the privileged parameters and its chosen primitive lies in the feasible recovery set, which converts “the oracle guessed” into “physically grounded.” The planner is then trained in two stages, supervised fine-tuning

on the filtered data followed by an embodied preference-optimization stage [26] that scores rollouts by their physical outcome.

We implement and evaluate Kino-VLA in high-fidelity simulation on a physically simulated Unitree Go2 in Isaac Lab [27], [28]. On held-out semantic-intervention failures the trained planner attains an attribution accuracy of 0.97, against 0.24 for a tuned rule-based state machine, and every sampled output parses against the action schema; the safety shield solves its quadratic program in under 0.2 ms at the ninety-ninth percentile; and the privileged-distillation extractor regresses friction, payload, actuator effort, and support state to within per-operator tolerances. Sim-to-real transfer is treated at the level of design (an actuator-network model and contact-signal randomization) and left to future hardware studies.

This paper makes the following contributions.

- We formalize the *recoverability dichotomy* as a falsifiable question and instantiate it as Kino-Fail, a parameterized failure-operator benchmark with privileged ground truth and three constructively matched attribution-ambiguity pairs that render proprioception-only recovery provably insufficient.
- We cast the safety layer as a CBF quadratic program that arbitrates the unconstrained output of a language model, with a forward-invariance guarantee in a capture-point reduced-order model and an explicit dual-rate latency budget under which the reflex loop keeps the robot alive while the planner reasons.
- We ground the cross-frequency interface in privileged physics: a distilled Kino-Tokens latent with a calibrated out-of-distribution score, an *active-sensing maneuver* that elicits an informative, deployment-consistent proprioceptive observation at the failure instant and doubles as a causal probe of the hidden cause, and a hindsight chain-of-thought pipeline whose truth-consistency filter rejects physically inconsistent annotations.
- We show, in simulation, that the resulting planner attributes and recovers from semantic-intervention failures where a tuned rule-based baseline cannot, evidence for the irreplaceability of vision-grounded reasoning on the B-class subspace.

II. RELATED WORK

A. Learned legged locomotion and online adaptation

Model-free reinforcement learning produces quadrupedal controllers that walk over challenging terrain [1], [2] and adapt online to unmodeled dynamics by encoding proprioceptive history into a latent that drives the policy [3], [4]. These controllers are trained at scale with massively parallel simulation [27]–[29] and transferred to hardware with actuator-network models of motor and gearbox nonlinearity [30]. Most use teacher-student privileged distillation, in which a student trained on onboard observations regresses a teacher’s access to privileged simulator state [1], [18]; we reuse exactly this mechanism for Kino-Tokens. The boundary is deliberate. These

methods encode dynamics to drive a continuous control policy, and we make no claim to improve their adaptation. We distill the same privileged signal into a *semantic* embedding that supports discrete decisions, and we target the B-class subspace where continuous adaptation is insufficient by construction rather than merely suboptimal.

B. Vision-language-action models and embodied foundation models

Large multimodal models have been adapted into robot policies that map language and pixels to actions [5]–[8], [31]–[33], and into high-level reasoners that sequence skills through language [34]–[36]. Legged variants extend this recipe to navigation [11]–[13], and visually closed-loop systems re-plan from RGB observations as a task unfolds [9], [10]. In all of these, action is generated open-loop or closed only on vision at camera rate, and on legged platforms locomotion is invoked as a black-box skill that returns no dynamics feedback. Kino-VLA adds a 1 kHz proprioceptive channel into the planner and a barrier-certified compiler beneath it, so the planner both sees the failure in the dynamics and cannot issue an action that destabilizes the body.

C. Failure detection, attribution, and language-model recovery

A growing line of work uses language and vision models to explain and correct robot failures. REFLECT [14] summarizes multimodal robot experience into a hierarchical trace for failure explanation, AHA [15] trains a vision-language model to detect and reason over manipulation failures, and DoReMi [37] detects and recovers from plan-execution misalignment. In parallel, classical legged-robot pipelines detect slip and anomalies from proprioception [38], and introspective perception predicts when a vision system will fail [39]. The distinctions are several. Attribution in the language-model line is performed after the fact, from low-frequency vision or audio, on fixed-base manipulators, and is injected as discrete text; classical legged anomaly detection stops at raising an alarm. Kino-VLA attributes *online*, during a reflex-maintained safe interval, from a 1 kHz dynamics channel, and closes the loop through an expanded recovery-primitive library and a persistent map rather than terminating at an explanation or an alarm.

D. Safe control and safety filters

Control barrier functions provide quadratic-program safety filters with forward-invariance guarantees [23], [40], and legged balance has long been analyzed through reduced-order capture-point [24] and divergent-component-of-motion [41] models built on the linear inverted pendulum and zero-moment-point formulation [42]. Recent work layers barrier functions with model predictive control for legged robots [43], certifies learning-based controllers with predictive safety filters [44], and surveys the broader safe-learning program [45].

These methods certify a *known* controller against bounded disturbances. We instead place the CBF quadratic program downstream of an unconstrained, potentially hallucinated language-model output and project discrete recovery primitives onto a capture-point safe set, which addresses formal safety for language-model-in-the-loop legged control.

E. Semantic maps and open-vocabulary perception for navigation

Open-vocabulary scene representations build navigable maps by grounding vision into 3D with CLIP [46] and SAM [47] features [48]–[50], and self-supervised traversability estimation learns terrain costs from experience [51]. These approaches write traversability from vision (vision $\rightarrow$ map). Kino-VLA inverts the direction: a physical failure rewrites the visually derived label of its 3D region and propagates the correction across visually homogeneous neighbors (physics $\rightarrow$ map), so that stepping through a single broken patch of thin ice condemns the entire visually uniform sheet.

REFERENCES

- [1] J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, “Learning quadrupedal locomotion over challenging terrain,” *Science Robotics*, vol. 5, no. 47, p. eabc5986, 2020.
- [2] T. Miki, J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, “Learning robust perceptive locomotion for quadrupedal robots in the wild,” *Science Robotics*, vol. 7, no. 62, p. eabk2822, 2022.
- [3] A. Kumar, Z. Fu, D. Pathak, and J. Malik, “RMA: Rapid motor adaptation for legged robots,” in *Robotics: Science and Systems (RSS)*, 2021.
- [4] I. M. A. Nahrendra, B. Yu, and H. Myung, “DreamWaQ: Learning robust quadrupedal locomotion with implicit terrain imagination via deep reinforcement learning,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [5] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, X. Chen, K. Chormanski, T. Ding, D. Driess *et al.*, “RT-2: Vision-language-action models transfer web knowledge to robotic control,” in *Conference on Robot Learning (CoRL)*, 2023.
- [6] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi, Q. Vuong, T. Kollar, B. Burchfiel, R. Tedrake, D. Sadigh, S. Levine, P. Liang, and C. Finn, “Open-VLA: An open-source vision-language-action model,” *arXiv preprint arXiv:2406.09246*, 2024.
- [7] Octo Model Team, D. Ghosh, H. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, T. Kreiman, C. Xu *et al.*, “Octo: An open-source generalist robot policy,” in *Robotics: Science and Systems (RSS)*, 2024.
- [8] D. Driess, F. Xia, M. S. M. Sajjadi, C. Lynch, A. Chowdhery, B. Ichter, A. Wahid, J. Tompson, Q. Vuong, T. Yu *et al.*, “PaLM-E: An embodied multimodal language model,” in *International Conference on Machine Learning (ICML)*, 2023.
- [9] Q. Bu, J. Zeng, L. Chen, Y. Yang, G. Zhou, J. Yan, P. Luo, H. Cui, Y. Ma, and H. Li, “Closed-loop visuomotor control with generative expectation for robotic manipulation,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [10] P. Zhi, Z. Zhang, Y. Zhao, M. Han, Z. Zhang, Z. Li, Z. Jiao, B. Jia, and S. Huang, “Closed-loop open-vocabulary mobile manipulation with GPT-4V,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
- [11] A.-C. Cheng, Y. Ji, Z. Yang, Z. Gongye, X. Zou, J. Kautz, E. Bryk, H. Yin, S. Liu, and X. Wang, “NaVILA: Legged robot vision-language-action model for navigation,” *arXiv preprint arXiv:2412.04453*, 2024.
- [12] Y. Mei, Y. Wang, S. Zheng, and Q. Jin, “QuadrupedGPT: Towards a versatile quadruped agent in open-ended worlds,” *arXiv preprint arXiv:2406.16578*, 2024.
- [13] Q. Wu, Z. Fu, X. Cheng, X. Wang, and C. Finn, “Helpful DoggyBot: Open-world object fetching using legged robots and vision-language models,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

- [14] Z. Liu, A. Bahety, and S. Song, "REFLECT: Summarizing robot experiences for failure explanation and correction," in *Conference on Robot Learning (CoRL)*, 2023.
- [15] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang, S. Tian, W. Yuan, R. Krishna, D. Fox, A. Mandlekar, and Y. Guo, "AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation," in *International Conference on Learning Representations (ICLR)*, 2025.
- [16] A. Jaegle, F. Gimeno, A. Brock, O. Vinyals, A. Zisserman, and J. Carreira, "Perceiver: General perception with iterative attention," in *International Conference on Machine Learning (ICML)*, 2021.
- [17] J.-B. Alayrac, J. Donahue, P. Luc, A. Miech, I. Barr, Y. Hasson, K. Lenc, A. Mensch, K. Millican, M. Reynolds *et al.*, "Flamingo: a visual language model for few-shot learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [18] D. Chen, B. Zhou, V. Koltun, and P. Krähenbühl, "Learning by cheating," in *Conference on Robot Learning (CoRL)*, 2020.
- [19] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," *arXiv preprint arXiv:1807.03748*, 2018.
- [20] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "LoRA: Low-rank adaptation of large language models," in *International Conference on Learning Representations (ICLR)*, 2022.
- [21] S. Bai, Y. Cai, R. Chen *et al.*, "Qwen3-v1 technical report," *arXiv preprint arXiv:2511.21631*, 2025.
- [22] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge, Y. Fan, K. Dang, M. Du, X. Ren, R. Men, D. Liu, C. Zhou, J. Zhou, and J. Lin, "Qwen2-v1: Enhancing vision-language model's perception of the world at any resolution," *arXiv preprint arXiv:2409.12191*, 2024.
- [23] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs for safety critical systems," *IEEE Transactions on Automatic Control*, vol. 62, no. 8, pp. 3861–3876, 2017.
- [24] J. Pratt, J. Carff, S. Drakunov, and A. Goswami, "Capture point: A step toward humanoid push recovery," in *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2006, pp. 200–207.
- [25] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- [26] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, and C. Finn, "Direct preference optimization: Your language model is secretly a reward model," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [27] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, "Isaac gym: High performance GPU-based physics simulation for robot learning," *arXiv preprint arXiv:2108.10470*, 2021.
- [28] M. Mittal, C. Yu, Q. Yu, J. Liu, N. Rudin, D. Hoeller, J. L. Yuan, R. Singh, Y. Guo, H. Mazhar, A. Mandlekar, B. Babich, G. State, M. Hutter, and A. Garg, "Orbit: A unified simulation framework for interactive robot learning environments," *IEEE Robotics and Automation Letters*, vol. 8, no. 6, pp. 3740–3747, 2023.
- [29] N. Rudin, D. Hoeller, P. Reist, and M. Hutter, "Learning to walk in minutes using massively parallel deep reinforcement learning," in *Conference on Robot Learning (CoRL)*, 2022.
- [30] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun, and M. Hutter, "Learning agile and dynamic motor skills for legged robots," *Science Robotics*, vol. 4, no. 26, p. eaau5872, 2019.
- [31] A. Brohan, N. Brown, J. Carbajal, Y. Chebotar, J. Dabis, C. Finn, K. Gopalakrishnan, K. Hausman, A. Herzog *et al.*, "RT-1: Robotics transformer for real-world control at scale," in *Robotics: Science and Systems (RSS)*, 2023.
- [32] X. Li, M. Liu, H. Zhang, C. Yu, J. Xu, H. Wu, C. Cheang, Y. Jing, W. Zhang, H. Liu, H. Li, and T. Kong, "Vision-language foundation models as effective robot imitators," in *International Conference on Learning Representations (ICLR)*, 2024.
- [33] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai, L. Groom, K. Hausman, B. Ichter *et al.*, " π_0 : A vision-language-action flow model for general robot control," 2024, physical Intelligence technical report.
- [34] M. Ahn, A. Brohan, N. Brown, Y. Chebotar, O. Cortes, B. David, C. Finn, C. Fu, K. Gopalakrishnan, K. Hausman *et al.*, "Do as i can, not as i say: Grounding language in robotic affordances," in *Conference on Robot Learning (CoRL)*, 2022.
- [35] J. Liang, W. Huang, F. Xia, P. Xu, K. Hausman, B. Ichter, P. Florence, and A. Zeng, "Code as policies: Language model programs for embodied control," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 9493–9500.
- [36] W. Huang, F. Xia, T. Xiao, H. Chan, J. Liang, P. Florence, A. Zeng, J. Tompson, I. Mordatch, Y. Chebotar, P. Sermanet, N. Brown, T. Jackson, L. Luu, S. Levine, K. Hausman, and B. Ichter, "Inner monologue: Embodied reasoning through planning with language models," in *Conference on Robot Learning (CoRL)*, 2022.
- [37] Y. Guo, Y.-J. Wang, L. Zha, Z. Jiang, and J. Chen, "DoReMi: Grounding language model by detecting and recovering from plan-execution misalignment," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
- [38] C. Yan, J. Qin, Q. Liu, and Q. Ma, "Slip detection and recovery for quadruped robots via orthogonal decomposition," *IEEE Transactions on Industrial Electronics*, vol. 72, no. 6, pp. 6166–6174, 2025.
- [39] S. Daftry, S. Zeng, J. A. Bagnell, and M. Hebert, "Introspective perception: Learning to predict failures in vision systems," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016.
- [40] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," in *European Control Conference (ECC)*, 2019, pp. 3420–3431.
- [41] J. Engelsberger, C. Ott, and A. Albu-Schäffer, "Three-dimensional bipedal walking control based on divergent component of motion," *IEEE Transactions on Robotics*, vol. 31, no. 2, pp. 355–368, 2015.
- [42] S. Kajita, F. Kanehiro, K. Kaneko, K. Fujiwara, K. Harada, K. Yokoi, and H. Hirukawa, "Biped walking pattern generation by using preview control of zero-moment point," in *IEEE International Conference on Robotics and Automation (ICRA)*, vol. 2, 2003, pp. 1620–1626.
- [43] R. Grandia, A. J. Taylor, A. D. Ames, and M. Hutter, "Multi-layered safety for legged robots via control barrier functions and model predictive control," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 8352–8358.
- [44] K. P. Wabersich and M. N. Zeilinger, "Linear model predictive safety certification for learning-based control," in *IEEE Conference on Decision and Control (CDC)*, 2018, pp. 7130–7135.
- [45] L. Brunke, M. Greeff, A. W. Hall, Z. Yuan, S. Zhou, J. Panerati, and A. P. Schoellig, "Safe learning in robotics: From learning-based control to safe reinforcement learning," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 5, pp. 411–444, 2022.
- [46] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning (ICML)*, 2021.
- [47] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, and R. Girshick, "Segment anything," in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [48] C. Huang, O. Mees, A. Zeng, and W. Burgard, "Visual language maps for robot navigation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [49] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, "Voxposer: Composable 3d value maps for robotic manipulation with language models," in *Conference on Robot Learning (CoRL)*, 2023.
- [50] B. Chen, F. Xia, B. Ichter, K. Rao, K. Gopalakrishnan, M. S. Ryoo, A. Stone, and D. Kappler, "Open-vocabulary queryable scene representations for real world planning," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [51] J. Frey, M. Mattamala, N. Chebrolu, C. Cadena, M. Fallon, and M. Hutter, "Fast traversability estimation for wild visual navigation," in *Robotics: Science and Systems (RSS)*, 2023.